

NodeFlow(On-site)

Sensing System

A guide to building your own soil sensor code

From a blank computer to readings on the screen

A research and outreach project of
UC Agriculture and Natural Resources and UC Santa Cruz

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Una versión en español de esta guía estará disponible próximamente.

A Spanish translation of this guide is on its way.

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1 Before you start

This guide takes you from a computer with nothing installed to a working soil sensor showing readings on a small screen. You do not need to know how to program. You will not write a single line of code.

1.1 What you need

- An Arduino Uno R3 board.
- A 16 by 2 LCD keypad shield, the display that presses onto the board.
- At least one sensor. The system reads three kinds, listed in section 3.2.
- A USB cable that carries data, not only power. If your computer does not notice the board when you plug it in, the cable is the first thing to suspect.
- A computer running Windows, macOS or Linux.
- Power. USB is enough on a bench. For a field, read section 4.

1.2 How long it takes

About twenty minutes the first time, most of which is installing the Arduino software. After that, generating a new file and uploading it takes two or three minutes.

2 Installing the Arduino IDE

The Arduino IDE is the free program that copies your file onto the board. You install it once.

1. Open <https://www.arduino.cc/en/software> in a web browser.
2. Download the version for your operating system. On Windows choose the installer rather than the app from the Microsoft Store, because the installer also sets up the drivers the board needs.
3. Run the file you downloaded and accept the default options. On Windows it will ask permission to install device drivers. Say yes to each one. On macOS, drag the Arduino icon into your Applications folder.
4. Open the Arduino IDE once, so it can finish setting itself up.

On macOS the first time you open it you may see a message saying the app cannot be checked for malicious software. Open System Settings, go to Privacy and Security, scroll down, and click Open Anyway.

2.1 Checking the board is seen

Plug the Arduino into your computer with the USB cable. A small light on the board should come on.

In the Arduino IDE:

1. Open the **Tools** menu.
2. Point at **Board** and choose **Arduino Uno**.
3. Point at **Port**. You should see an entry that appeared when you plugged the board in. On Windows it looks like **COM3**. On macOS it looks like **/dev/cu.usbmodem** followed by some numbers. Choose it.

If no new port appears, work through these in order: try a different USB cable, try a different USB socket, and on Windows check that the driver installed. A cable that only carries power is by far the most common cause.

3 Wiring your sensors

3.1 Which pins you can use

The display shield covers most of the board. What it leaves you is:

- **A1 to A5** for sensors. Five ports.
- **A0** is taken by the buttons on the shield. Do not use it.
- **D2 and D3** are used when a sensor is wired without an adapter, described in section 3.3.

3.2 The sensors this system reads

Capacitive soil moisture probe

A low cost probe, such as the DFRobot SEN0308, with three wires. Red goes to 5V, black goes to GND, and the yellow signal wire goes to one of A1 to A5. It needs no adapter.

This probe reads *high in dry soil and low in wet soil*. That is a property of the probe, and the code takes it into account.

Irrrometer Watermark 200SS

Measures how hard a plant has to pull to get water out of the soil, which is called soil water tension and is reported in kPa. A low number means wet soil. A high number means dry soil.

There are two ways to connect it, and they are genuinely different:

- **Through a 200SS-VA3 adapter.** The adapter does the difficult electrical work and hands the board a simple voltage. One adapter carries up to three Watermarks and one temperature probe on a single port. This is the recommended way, and the way to fit more than five sensors.
- **Wired straight to the board.** One sensor only. See section 3.3 before you try this.

Irrrometer 200TS soil temperature sensor

Measures soil temperature. It can sit on its own, or share a 200SS-VA3 adapter with Watermark sensors.

3.3 Wiring a sensor without the adapter

The form lets you wire a Watermark or a temperature sensor straight to the board. You will need a 10 kilohm resistor for each one. The sensor goes between a digital pin and your chosen analog pin, and the resistor goes from that same analog pin to a second digital pin. The code uses D2 and D3 for this.

One sensor only. Wet soil conducts electricity between two bare sensors, so each one ends up partly reading the other, and over time the metal inside them corrodes. This is the manufacturer's own warning. If you want more than one, use a 200SS-VA3 adapter.

The combined Watermark and temperature option needs the adapter. It exists because the adapter is what joins the two sensors together. The form does not offer direct wiring for it.

4 Powering the board

On a bench the USB cable powers everything. In a field it does not, so this section is about what to use instead.

4.1 How much current the system draws

Part	Roughly
Arduino Uno, board alone	45 mA
Display backlight	25 mA
Display logic	2 mA
200SS-VA3 adapter	1.5 mA
Capacitive probe	5 mA
Typical total, backlight on	75 mA
Typical total, backlight off	50 mA

Only the adapter figure comes from a datasheet. The rest are typical values for this hardware and vary between boards and shields. Measure yours if runtime matters: put a multimeter in series with the battery and read the current with the backlight on and off.

The backlight is a third of the load. That is why the screen goes dark after five minutes without a button press, and why any button brings it straight back. Leaving it on shortens a battery by roughly a third.

4.2 Two ways to get power in

The barrel socket is the round connector. It expects 7 to 12 volts and goes through a regulator on the board that turns the surplus into heat. At 9 volts in and 5 volts out, roughly four tenths of the battery's energy is wasted warming the regulator. It works, and it is what the low battery warning watches.

The USB socket expects exactly 5 volts and skips the regulator, so nothing is wasted. It is the efficient option, but the low battery warning cannot see anything through it.

4.3 The options

Battery	Where	Capacity	Rough runtime
9 V alkaline (PP3)	barrel	550 mAh	7 to 11 hours
6 AA alkaline	barrel	2500 mAh	30 to 50 hours
8 AA alkaline	barrel	2500 mAh	30 to 50 hours
USB power bank	USB	10000 mAh	5 to 8 days
12 V sealed lead acid	barrel	7000 mAh	3 to 6 weeks

Runtimes assume the backlight is mostly off. Halve them if it stays on.

9 volt alkaline

The obvious choice, and the worst one. A PP3 holds very little energy for its size, and the regulator throws away nearly half of what it does hold. Fine for an afternoon of testing. Do not plan a season around it.

AA cells in a holder

Six AA cells give about 9 volts, which suits the barrel socket. This is the best simple choice for a few days of logging: AA cells hold four or five times the energy of a PP3 and cost less.

Eight cells give 12 volts, which lasts no longer in practice, because the extra three volts are turned into heat rather than reading time.

Four AA cells give 6 volts, which is below what the barrel socket needs. The board will run unevenly or not at all. Use six.

USB power bank

The most energy for the money, and no regulator waste. Two things to know:

- Many power banks switch themselves off when the load is small, and 50 mA is small. If yours does this, look for one sold as suitable for low power devices, or one with a switch that keeps it awake.
- **The low battery warning does not work on USB power.** The board cannot tell the difference between a full bank and one that is about to cut out. Do not rely on the warning when running this way.

Sealed lead acid, with or without solar

For something left in a field. A 12 volt 7 Ah battery runs the system for weeks and takes a small solar panel and charge controller happily.

Because the regulator wastes the difference between 12 volts and 5, a separate switching regulator set to 5 volts, feeding the 5V pin rather than the barrel socket, roughly doubles the runtime. That is a wiring change rather than something the code knows about.

4.4 The low battery warning

The board measures its own supply against a fixed reference inside the chip. When a battery on the barrel socket runs down far enough that the regulator can no longer hold 5 volts, the screen shows a warning on the second line.

This only works on battery power through the barrel socket. On USB, including a USB power bank, the supply stays at 5 volts until the moment it disappears, so no warning can be given. This is a limit of the hardware, not of the code.

5 Using the form

Open the code generator in a web browser:

<https://fernlag.github.io/NodeFlow-Onsite-Sensing-System/>

Once the page has loaded once, it keeps working without an internet connection.

There is a language button in the top right corner. It switches the whole page between English and Spanish, and remembers your choice.

5.1 One block per sensor

The page starts with one block. Each block describes one sensor plugged into one port. Use the **Add another** button if you have more than one sensor.

For each block you answer four questions:

1. **Sensor type.** Which sensor this is.
2. **Port.** Which port you plugged it into.
3. **What you want to see on the screen.** The measurement, described in section 8.
4. **How it should appear.** A number, a percentage, a bar, or words describing the state.

5.2 The settings underneath

Below those four questions is a section of settings. It only shows the ones that matter for the choices you made. If you pick a measurement that needs nothing configured, the section disappears entirely. This is deliberate: a setting you should not touch is not shown.

Each setting has a small **i** button beside it. Press it, or move onto it with the Tab key, and it explains what the number means and where to get it.

The default values are starting points, not measurements of your soil. A default is a guess. The two calibration values for a capacitive probe in particular must be measured, and section 9 explains how.

5.3 Generating the file

Press the large **Generate** button at the bottom. A window appears asking for your name, country and email, and for a name for the file.

This information goes to the research project. The consent form is linked in that window, and you have to tick the box before the button becomes available. Nothing is sent until you press confirm. The privacy policy on the site explains exactly what is recorded and how to have it removed.

If you tick the date and time option, the file name gets a stamp on the end, so each download sits alongside the last one instead of being numbered by your browser.

The file downloads to wherever your browser puts downloads. It ends in **.ino**, which is the Arduino file type.

6 Putting the file on the board

1. Find the downloaded file and open it. The Arduino IDE starts.
2. The IDE offers to move the file into a folder of the same name. Say yes. That is simply how Arduino files are stored.
3. Check **Tools** → **Board** still says Arduino Uno, and **Tools** → **Port** still points at your board.
4. Press the arrow button in the top left to upload. Lights on the board flicker while it copies.
5. When the bar at the bottom says **Done uploading**, the display starts showing readings.

Nothing needs installing for the code to work. It uses only the display library that comes with the Arduino IDE.

7 Reading the screen

The top line shows the port, a short sensor name, and the value.

```
A2 WM200SS 34kPa
OK irrigate
```

The bottom line shows whatever you chose: a bar, a percentage, or words. Press the **left** and **right** buttons on the shield to step through your sensors.

The backlight switches off after five minutes without a button press, to save power. Any button wakes it.

7.1 Which way the numbers run

This is worth reading once, because it prevents a common misunderstanding.

Every percentage runs from dry to wet. Zero means dry. One hundred means wet. This is true for every percentage the system shows, without exception.

Tension in kPa runs the other way, because that is what tension is. Zero kPa is soaking wet soil. A high number is dry soil. So that this cannot be confused, the second line of the display always spells out the state next to the number, as in the example above.

Raw readings report the sensor itself. A capacitive probe reads high in dry soil, so a raw reading from it goes up as the soil dries. If you want a number that goes up as soil gets wetter, choose the percentage rather than the raw value.

8 The measurements you can choose

Raw value Exactly what the sensor put on the pin, with nothing done to it. Useful for calibration and for checking a sensor is alive.

Raw value as a percentage The raw reading placed on a scale between your dry and wet calibration points. Zero is dry, one hundred is wet.

Management thresholds Three plain states rather than a number: too dry, about right, or saturated. For a Watermark you choose your soil texture and the two tension thresholds fill themselves in.

Tension in kPa Soil water tension. Low is wet, high is dry.

Temperature Soil temperature, in Fahrenheit or Celsius.

Wetting front Whether irrigation water has reached a given depth yet. Needs two probes at different depths, so it needs two blocks.

Vertical flow rate How fast water travelled between the two depths, in centimetres per hour.
Also needs two probes.

Volumetric soil moisture How much of the soil's volume is water, as a percentage.

Total available water How much of the water the soil can hold is actually available to a crop.
Zero at the wilting point, one hundred at field capacity.

9 Calibrating a capacitive probe

A capacitive probe needs two numbers before a percentage means anything: what it reads in air, and what it reads in water. They differ between probes, so measure yours.

1. In the form, choose **Raw value** as the measurement for that sensor, and **Raw value** as the display. Generate the file and upload it.
2. Hold the probe in open air, away from your hand. Write down the number on the screen. It will be somewhere near 590.
3. Stand the probe in a glass of water, up to but not past the line marked on it. Write down that number. It will be somewhere near 280.
4. Go back to the form, choose the measurement you actually want, and put your two numbers into the air value and water value boxes.
5. Generate the file again and upload it.

Do not put the probe in water past the marked line. The electronics at the top are not waterproof.

10 When something looks wrong

The screen is blank, or shows a row of solid blocks Turn the small screw on the shield, which is the contrast control. A new shield is often set to one extreme or the other.

A port says No Sensor Nothing is detected on that pin. Check both ends of the wiring. If the sensor goes through a 200SS-VA3 adapter, give it time: the adapter updates each sensor once every five minutes, so a reading can be up to five minutes old.

The percentage seems backwards Your air and water calibration numbers are probably the wrong way round. See section 9.

A battery warning appears while on USB Ignore it. The check only means something on battery power through the barrel socket. See section 4.

The board keeps switching off in the field If it runs from a USB power bank, the bank is probably shutting down because the load is too small for it to notice. See section 4.

The numbers do not match my soil The calibration values and the soil thresholds belong to your field, not to us. Change them in the form and generate the file again.

The upload fails Check the port is still selected. Unplug the board, plug it back in, select the port again and retry. If it still fails, close the Serial Monitor window if you have it open, as it can hold the port.

11 Working without an internet connection

The page works offline once it has been opened online at least once. Fill in the form and download files with no connection at all.

Clearing your browser's history or site data for this page removes the offline copy, along with your saved details. Open the page once more while online and it comes back.

12 Getting help

The contact address is at the bottom of every page on the site. Tell us which sensor, which port, which measurement, and what the screen shows. If you can, include the file name you generated: it lets us find your exact configuration.

The generated code is provided as is, without warranty. Before you irrigate on the strength of a reading, check that it matches what the soil is actually doing: dig, or compare it against an instrument you already trust.